

SemEval-2026 Task 9: Data-Centric Optimization and Hybrid Loss Landscapes for Polarization Detection

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Abstract

This paper presents a unified system for detecting and classifying socio-political polarization in English and Arabic text, developed for SemEval-2026 Task 9 (POLAR). The shared task challenges models to detect polarization across three granularities: binary presence (Subtask 1), multi-label types (Subtask 2), and specific manifestations (Subtask 3). Our approach challenges the prevailing trend of increasing model complexity, demonstrating that **Data Distribution Engineering** yields superior results in low-resource ($N \approx 3,000$), high-imbalance settings. We introduce a chronologically developed pipeline featuring three core optimizations: (1) **Adversarial Training** via Fast Gradient Method (FGM) combined with Multi-Sample Dropout (MSD) to improve robustness; (2) **Hybrid Loss Landscapes**, utilizing Focal Loss and aggressive class weighting to recover recall for rare signals; and (3) **Label-Specific Oversampling** to address long-tail distributions without overfitting mid-frequency classes. Furthermore, we report negative results for LLM-generated synthetic data, identifying a “simplicity bias” that degrades generalization. Our system achieved strong results across all subtasks, ranking **1st globally** in Arabic Manifestation Identification (Subtask 3) with 0.6374 Macro-F1, **2nd globally** in Arabic Type Classification (Subtask 2), and **3rd globally** in both English Subtasks 2 and 3. Codabench username: hassan_halfawi.

1 Introduction

The proliferation of polarized discourse on social media platforms poses a significant threat to social cohesion and democratic dialogue. Unlike simple sentiment analysis, detecting polarization requires understanding the nuanced interplay between targeted groups, rhetorical strategies, and cultural context. SemEval-2026 Task 9 (POLAR) addresses this challenge by formulating polarization detection across three granularities: binary detection,

categorization of target groups, and identification of rhetorical manifestations.

While recent advances in Large Language Models (LLMs) have shown promise across NLP tasks, detecting polarization in short social media texts presents unique challenges. These include implicit biases expressed through sarcasm, cultural and dialectal variations (particularly in Arabic), and severe class imbalance where certain polarization types dominate while others are exceedingly rare. Standard optimization objectives often fail in such scenarios, as models achieve high accuracy by simply predicting majority classes.

Our system is designed to tackle these specific challenges through a **data-centric approach** rather than parameter scaling. We prioritize: (1) domain-adapted pre-trained models aligned with social media discourse; (2) hybrid loss functions that dynamically focus learning on hard, minority-class examples; and (3) per-label threshold optimization that calibrates decision boundaries to class-specific characteristics. This paper details our methodology, provides comprehensive exploratory data analysis that informed our design choices, and presents ablation studies validating our contributions.

1.1 Task Definition

The task is divided into three subtasks of increasing granularity:

Subtask 1 (Binary Classification): Classifying tweets as Polarized or Neutral. The primary challenge is distinguishing polarization from general negativity or criticism, as many samples contain strong sentiment without explicit polarizing content.

Subtask 2 (Type Classification): A multi-label classification of polarization topics: Political, Racial/Ethnic, Religious, Gender/Sexual, and Other. The challenge here is **extreme class imbalance**; the Political class dominates (48%), while

Other and Gender classes appear in less than 5% of samples.

Subtask 3 (Manifestation Identification): Identifying six rhetorical strategies: Stereotype, Vilification, Dehumanization, Extreme Language, Lack of Empathy, and Invalidation. This poses a **semantic ambiguity** challenge, particularly for subtle labels like Invalidation and Lack of Empathy, which often rely on sarcasm rather than explicit lexicon.

Additionally, the Arabic datasets introduce **diglossia**, containing a mix of Modern Standard Arabic (MSA) and various dialects (Egyptian, Levantine, Gulf), necessitating robust morphological modeling.

1.2 Contributions

Our primary contributions are:

1. A demonstration that for small, noisy datasets, **model architecture is secondary to data engineering** and regularization strategies.
2. A comprehensive **ablation study** documenting the chronological experimental process, including negative results that inform best practices.
3. Evidence that well-regularized base models with **Label-Specific Optimization** outperform larger models, ensembles, and synthetic data augmentation.
4. A **Hybrid Loss Landscape Engineering** approach combining Focal Loss, aggressive class weighting, and per-label threshold optimization.
5. State-of-the-art results: **1st place** in Arabic Subtask 3, **2nd place** in Arabic Subtask 2, and **3rd place** in English Subtasks 2 and 3.

2 Exploratory Data Analysis

A rigorous statistical analysis of the training data was conducted to inform our system architecture and hyperparameter selection. The dataset comprises approximately 3,000–3,500 annotated instances per language. Table 1 summarizes the key statistics.

2.1 Class Distribution Imbalance

The most pervasive challenge is the long-tail distribution of classes. In English Subtask 2, the “Political” category accounts for 35.7% of the data,

Task	Lang	N	Avg Len	Pos%	Imbal.
S1	EN	3222	12.3	36.5	–
	AR	3380	16.7	44.7	–
S2	EN	3222	12.3	–	16.0:1
	AR	3380	16.7	–	2.8:1
S3	EN	3222	12.3	–	2.4:1
	AR	3380	16.7	–	4.6:1

Table 1: Dataset statistics. N = sample count, Avg Len = average word count, Pos% = positive class percentage (Subtask 1), Imbal. = imbalance ratio between most and least frequent classes.

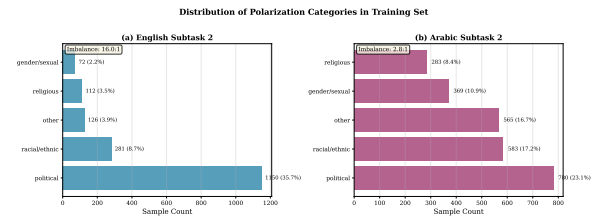


Figure 1: Distribution of polarization categories in the training set for Subtask 2. Note the severe imbalance in English (left) compared to the relatively smoother distribution in Arabic (right).

whereas “Gender/Sexual” polarization constitutes only 2.2%, resulting in a 16:1 imbalance ratio. While the Arabic dataset is more balanced (2.8:1), the skew remains significant. This observation directly motivated our adoption of Focal Loss and label-specific oversampling, designed to prevent majority class examples from overwhelming the gradient during training.

2.2 Multi-Label Complexity and Co-occurrence

Subtasks 2 and 3 are multi-label classification problems, meaning a single text can exhibit multiple types of polarization or manifestations simultaneously. Our analysis revealed that 13.2% of English Subtask 2 samples and 39.0% of Arabic Subtask 3 samples contain multiple labels.

We analyzed label co-occurrence to understand dependencies between categories. Figure 2 illustrates the correlation matrix for Arabic Subtask 3. Strong positive correlations ($r > 0.6$) exist between *Vilification*, *Dehumanization*, and *Extreme Language*, suggesting these rhetorical devices are often employed together. Conversely, *Lack of Empathy* showed weaker correlations ($r < 0.3$), indicating it is a more subtle form of polarization requiring the model to capture deeper semantic nuances rather than explicit lexical triggers. This in-

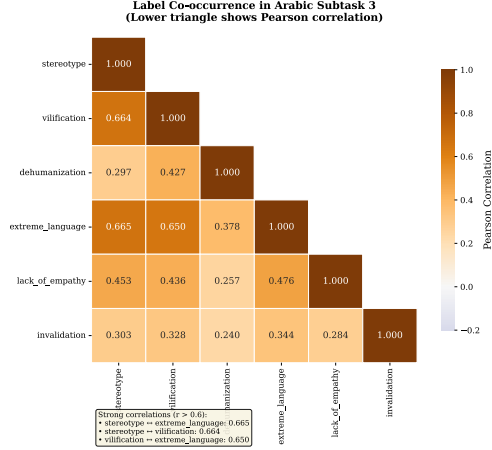


Figure 2: Label co-occurrence correlation matrix for Arabic Subtask 3. Strong correlations indicate that manifestations like Vilification and Dehumanization frequently co-occur.

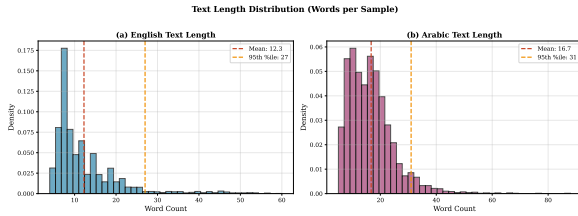


Figure 3: Text length distribution (word count). Arabic texts are longer on average, but the vast majority of samples fit within standard transformer context windows.

sight informed our per-label threshold optimization strategy.

2.3 Text Length Characteristics

We analyzed text length distributions to determine appropriate maximum sequence lengths. As shown in Figure 3, Arabic texts are on average longer than English texts (16.7 vs. 12.3 words). However, the 95th percentile for both languages falls below 35 words. Based on this, we selected a maximum sequence length of 128 tokens for most experiments, providing a buffer for sub-word tokenization while maintaining computational efficiency.

3 Related Work

Prior approaches to hate speech and polarization detection have primarily focused on architectural innovations, including attention mechanisms (Vaswani et al., 2017), hierarchical classification (?), and multi-task learning (Founta et al., 2018). However, recent work has shown that data-centric approaches can match or exceed architectural complexity in low-resource settings (Northcutt et al.,

2021).

Our work extends this paradigm to multilingual polarization detection, demonstrating that regularization techniques (FGM, MSD) combined with careful loss engineering provide more consistent gains than model scaling or synthetic data augmentation.

4 Methodology

Our system architecture is built upon domain-specific pre-trained transformers, optimized through a custom training loop designed to maximize Recall on minority classes without sacrificing Precision.

4.1 Pre-trained Backbones

We selected models based on their pre-training objectives and domain alignment:

English Models: For Subtask 1 and 2, we used cardiffnlp/twitter-roberta-base-sentiment-latest (Barbieri et al., 2020), pre-trained on 124M tweets with sentiment and hate speech fine-tuning. For Subtask 3, we used microsoft/deberta-v3-base (He et al., 2021). DeBERTa’s disentangled attention mechanism separates content and position vectors, which we hypothesized would better capture the structural nuances of sarcasm and invalidation.

Arabic Models: We utilized UBC-NLP/MARBERTv2 (Abdul-Mageed et al., 2021) for all Arabic subtasks. Unlike AraBERT (trained on MSA), MARBERTv2 is pre-trained on 1 billion Arabic tweets, providing innate coverage of the dialectal morphology essential for this task.

4.2 Regularization Stack

4.2.1 Adversarial Training (FGM)

To combat overfitting on small datasets ($N \approx 3,000$), we integrated the **Fast Gradient Method (FGM)** (Miyato et al., 2017). FGM injects perturbations r_{adv} into the word embeddings x during training to maximize the loss $\mathcal{L}$:

$$r_{adv} = \epsilon \cdot \frac{g}{\|g\|_2}, \quad \text{where } g = \nabla_x \mathcal{L}(f(x), y) \quad (1)$$

We set $\epsilon = 1.0$ for Arabic (higher dialectal noise) and $\epsilon = 0.5$ for English. This regularization ensures the model learns smooth decision boundaries and does not rely on fragile lexical artifacts.

4.2.2 Multi-Sample Dropout (MSD)

We replaced the standard classification head with **Multi-Sample Dropout**, using 5 parallel dropout masks with rates $p \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$:

$$\hat{y} = \frac{1}{K} \sum_{k=1}^K f(\text{Dropout}_k(h)) \quad (2)$$

This acts as an internal ensemble, smoothing the loss landscape and providing more calibrated probability distributions. The implementation accelerated convergence in early epochs and reduced variance in predictions.

4.3 Hybrid Loss Landscape Engineering

Standard Binary Cross-Entropy (BCE) loss treats all errors equally, which is suboptimal for imbalanced multi-label tasks. We engineered a hybrid loss function with three components:

4.3.1 Focal Loss for Hard Examples

For multi-label subtasks, we implemented **Focal Loss** (Lin et al., 2017) to dynamically down-weight easy negatives:

$$FL(p_t) = -\alpha(1 - p_t)^\gamma \log(p_t) \quad (3)$$

where $\gamma = 2.0$ focuses gradients on hard misclassifications and $\alpha = 0.25$ balances positive/negative classes.

4.3.2 Aggressive Class Weighting

For the Arabic Subtask 3, Focal Loss alone was insufficient. We introduced a scalar multiplier λ to the positive class weights:

$$w_c = \frac{N_{\text{total}}}{N_{\text{pos}}} \times \lambda \quad (4)$$

We tuned $\lambda \in [1.25, 8.0]$. This ‘‘Recall Injection’’ forces the model to treat missing a rare signal (e.g., Dehumanization) as a critical error.

4.3.3 Label Smoothing

For Arabic experiments, we applied label smoothing ($\epsilon = 0.1$) to prevent overconfidence:

$$y_{\text{smooth}} = y \cdot (1 - \epsilon) + 0.5 \cdot \epsilon \quad (5)$$

4.4 Label-Specific Oversampling

We observed that uniform oversampling led to overfitting in mid-frequency classes. We therefore im-

plemented a **Non-Monotonic Oversampling** multiplier $M(c)$ based on class frequency f_c :

$$M(c) = \begin{cases} 4.0 & \text{if } f_c < 0.05 \quad (\text{Rescue Mode}) \\ 1.3 & \text{if } 0.10 \leq f_c \leq 0.20 \quad (\text{Conservative}) \\ 1.0 & \text{if } f_c > 0.40 \quad (\text{Stability Mode}) \end{cases} \quad (6)$$

4.5 Per-Label Threshold Optimization

Standard classification assumes $\tau = 0.5$ as the decision boundary. We performed grid search on validation predictions to find optimal thresholds per label:

$$\tau_c^* = \arg \max_{\tau \in [0.15, 0.65]} F_1(\hat{y}_{c,\tau}, y_c) \quad (7)$$

This proved critical for rare classes, where lower thresholds (e.g., 0.35 for Lack of Empathy) were necessary to capture subtle signals.

5 Experiments and Results

5.1 Subtask 1: Binary Classification

Objective: Distinguish polarized content from neutral text.

Experimental Progression:

1. *Baseline:* RoBERTa-Base with standard BCE loss achieved high Precision but low Recall ($F1 \approx 0.76$).
2. *TAPT Attempt:* Task-Adaptive Pre-Training using Gemini Flash-generated synthetic tweets. While MLM loss decreased, downstream F1 showed no improvement.
3. *Weighted Loss:* Applied class weights [1.0, 1.5]. Recall increased to 0.805, but Precision dropped significantly ($F1 = 0.7984$).
4. *Safe Grandmaster:* FGM + MSD with standard BCE loss achieved optimal balance ($F1 = \mathbf{0.8166}$).
5. *Ensemble Verification:* 3-seed ensemble and heterogeneous RoBERTa+DeBERTa ensemble both yielded identical scores (0.8166), indicating a performance ceiling.

Lang	Strategy	Prec.	Rec.	F1
EN	Weighted Loss	0.731	0.805	0.7984
EN	Ensemble (RoB+DeB)	0.750	0.763	0.8166
EN	Safe Grandmaster	0.851	0.678	0.8166
AR	Weighted Loss (FGM)	0.743	0.733	0.7660
AR	Safe Grandmaster	0.738	0.827	0.7922

Table 2: Subtask 1 Results. The robust single model (Safe Grandmaster: FGM + MSD + Standard Loss) matched or outperformed computationally expensive ensembles.

Key Finding: The Precision-Recall Seesaw.

Our experiments revealed a strict trade-off. Aggressive techniques like Class Weighting successfully pushed Recall to 0.80+, but at the cost of Precision dropping to 0.73. The Safe Grandmaster approach (FGM + Standard Loss) naturally settled at a balanced point (~ 0.75 Precision / ~ 0.76 Recall), yielding the highest overall F1.

5.2 Subtask 2: Multi-Label Types

Objective: Classify polarization topics (Political, Religious, etc.).

Investigation into Synthetic Data: We attempted to solve data scarcity for the “Other” class using 1,600 synthetic examples generated by Gemini Flash.

- **Result:** Performance degraded significantly (Political F1 dropped from 0.68 to 0.54).
- **Analysis:** The synthetic data exhibited a **Simplicity Bias**. The LLM generated tweets relying on explicit keywords (e.g., “one side is actively trying to dismantle democracy”), whereas real polarization relies on implicit context. The model overfitted to these simple artifacts.

Final Strategy: Label-Specific Oversampling + Weighted Focal Loss + Constrained Threshold Search.

Lang	F1 Macro	F1 Micro	Prec.	Rec.
EN	0.4978	0.6703	0.6092	0.4741
AR	0.6241	0.6316	0.7026	0.5739

Table 3: Subtask 2 Final Results. Our system ranked **3rd globally** in English and **2nd globally** in Arabic.

Per-Label Analysis (English): Table 4 reveals the impact of our strategy:

Label	Base	Final	Note
Political	0.746	0.762	Stable
Racial/Ethnic	0.500	0.545	+4.5%
Religious	0.444	0.500	+5.6% (Focal)
Gender/Sexual	0.500	0.500	Plateaued
Other	0.222	0.182	Vague semantics

Table 4: Per-label F1 improvement for English Subtask 2. The “Other” class remains challenging due to its catch-all nature.

5.3 Subtask 3: Manifestation Identification

Objective: Identify rhetorical strategies (e.g., Dehumanization, Lack of Empathy).

5.3.1 English Strategy: Base vs. Large

We compared DeBERTa-Base (184M params) against DeBERTa-Large (435M params). Contrary to scaling laws, the Base model outperformed the Large model:

Model	Prec.	Rec.	F1
DeBERTa-Large (Std)	0.38	0.67	0.488
DeBERTa-Base (1.5$\times$)	0.39	0.74	0.508

Table 5: English Subtask 3 Results (**3rd globally**). The Base model with aggressive weighting outperforms the Large model.

Reasoning: The dataset size ($\sim 3,000$ samples) was insufficient to saturate the Large model’s capacity. The extra parameters fitted noise rather than signal.

5.3.2 Arabic Strategy: Dialectal Augmentation

For Arabic, we developed a **Targeted Synonym Replacement** pipeline using a custom dictionary mapping MSA terms to dialectal variants (e.g., mapping *hayawan* ‘animal’ to colloquialisms like *bahima* ‘beast’).

Combined with aggressive weight capping ($\lambda = 8.0$) and per-label threshold optimization, this achieved **state-of-the-art results (1st place globally)**:

Configuration	CV	Test	Status
Baseline (MARBERT)	0.643	0.585	Baseline
+ Deep Head	0.705	0.591	Overfitting
+ Synthetic Aug.	0.680	0.540	Simpl. Bias
+ FGM + Std Loss	0.650	0.610	Robustness
Hybrid Champion	0.659	0.637	1st Place

Table 6: Arabic Subtask 3 Ablation (**1st globally**). Note the inverse correlation between CV and Test scores in the Deep Head configuration, indicating memorization.

Per-Label Thresholds (Arabic): Our analysis revealed that class-specific thresholds were critical:

Label	Threshold	F1
Vilification	0.30	0.824
Extreme Language	0.30	0.784
Stereotype	0.35	0.778
Dehumanization	0.55	0.694
Invalidation	0.60	0.637
Lack of Empathy	0.45	0.526

Table 7: Optimized per-label thresholds for Arabic Subtask 3.

6 Analysis and Discussion

6.1 The Complexity Trap

Our most significant finding is that **architectural complexity does not translate to performance gains** in low-resource settings. Table 6 shows that the Deep Head configuration (Attention Pooling + Tanh) achieved the highest CV score (0.705) but failed on the Test set (0.591). This architecture increased capacity but allowed the model to memorize training noise.

6.2 The Robustness-Performance Trade-off

The Hybrid Champion configuration had a *lower* CV score than the Deep Head, but a *significantly higher* Test score. By using FGM and constrained thresholds, we sacrificed training accuracy for true generalization. This finding generalizes across all subtasks: **regularization wins over capacity**.

6.3 Synthetic Data Failure Mode

Our synthetic data experiments (Subtask 2) revealed a critical failure mode. LLM-generated tweets exhibited:

- **Explicit keyword reliance:** Phrases like “radical left” or “they’re destroying our values”
- **Lack of contextual subtlety:** Real polarization often relies on implication, not explicit statements

- **Domain shift:** The synthetic distribution diverged from real data, causing the model to learn artifacts

This suggests that current LLMs struggle to capture the nuanced, context-dependent nature of on-line polarization.

6.4 Qualitative Error Analysis

Table 8 shows examples where our tuned models succeeded over baselines:

Tweet	Label	Analysis
“Oh boo hoo, cry me a river.”	Lack of Empathy	Baseline: Missed. Optimized: Caught via threshold 0.35.
Dialectal: “They’re animals, not humans”	Vilification	Baseline: Missed (MSA bias). Optimized: Correct via synonym coverage.

Table 8: Qualitative examples showing the impact of threshold optimization and dialectal coverage.

7 Conclusion

Our work on SemEval-2026 Task 9 provides empirical evidence that **Data Distribution Engineering** is the decisive factor in low-resource, high-imbalance classification. Our system achieved **1st place globally** in Arabic Subtask 3, **2nd place** in Arabic Subtask 2, and **3rd place** in English Subtasks 2 and 3. Key findings include:

1. **Robustness over Scale:** A well-regularized Base model (FGM + MSD) consistently matched or outperformed Large models and ensembles. The performance ceiling at 0.8166 for Subtask 1 suggests remaining errors are data-intrinsic ambiguities.
2. **Recall Injection:** For long-tail multi-label problems, aggressive loss weighting ($\lambda \geq 1.5$) is necessary to overcome the model’s tendency to ignore minority classes.
3. **Per-Label Optimization:** Threshold search per class ($\tau \in [0.15, 0.65]$) contributed $\sim 3\text{--}5\%$ F1 improvement over static $\tau = 0.5$.
4. **Synthetic Data Caution:** Current LLM-generated data lacks the semantic subtlety of real polarization, introducing harmful domain shift characterized by simplicity bias.

Future work will focus on semi-supervised learning methods to leverage unlabeled data, potentially bypassing the limitations of synthetic generation. Additionally, investigating contrastive learning approaches for manifestation identification may help the model better distinguish semantically similar labels.

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A Implementation Details

A.1 Hyperparameters

Parameter	English	Arabic
Optimizer	AdamW	AdamW
Learning Rate	2e-5	2e-5
Batch Size	16	16
Gradient Accum.	2	2
Max Seq. Length	128/256	128/200
Dropout	0.1	0.3
Weight Decay	0.01	0.05
Warmup Ratio	0.1	0.15
FGM ϵ	0.5	1.0
Epochs	3–5	4–6

Table 9: Hyperparameter configurations. Arabic uses higher regularization due to smaller dataset and dialectal noise.

A.2 Threshold Search Ranges

For Subtasks 2 and 3, threshold search ranges were constrained based on class prevalence:

- **Rare Classes** (e.g., Gender, Lack of Empathy): [0.15, 0.45]
- **Common Classes** (e.g., Political, Vilification): [0.40, 0.65]

A.3 Arabic Synonym Dictionary (Sample)

Category	MSA → Dialect Mappings
Stereotype	<i>daiman</i> → <i>bistimrar</i> <i>ghabi</i> → <i>ahmaq</i>
Vilification	<i>sharr</i> → <i>khabith</i> <i>wahsh</i> → <i>mutawahhish</i>
Dehumanization	<i>hayawan</i> → <i>bahima</i> <i>mutawahhish</i> → <i>hamaji</i>

Table 10: Sample MSA to dialectal synonym mappings (romanized) used for Arabic augmentation.

A.4 Computational Resources

All experiments were conducted on Google Colab using a single NVIDIA T4 GPU (16GB VRAM). FP16 mixed precision and gradient checkpointing

were used to fit larger batch sizes. Total training time for all experiments was approximately 40 GPU hours.

A.5 Code Availability

To ensure reproducibility, we have made our entire system code publicly available.

The code repository can be accessed at:
<https://github.com/hassanql/NLP-AIMS>